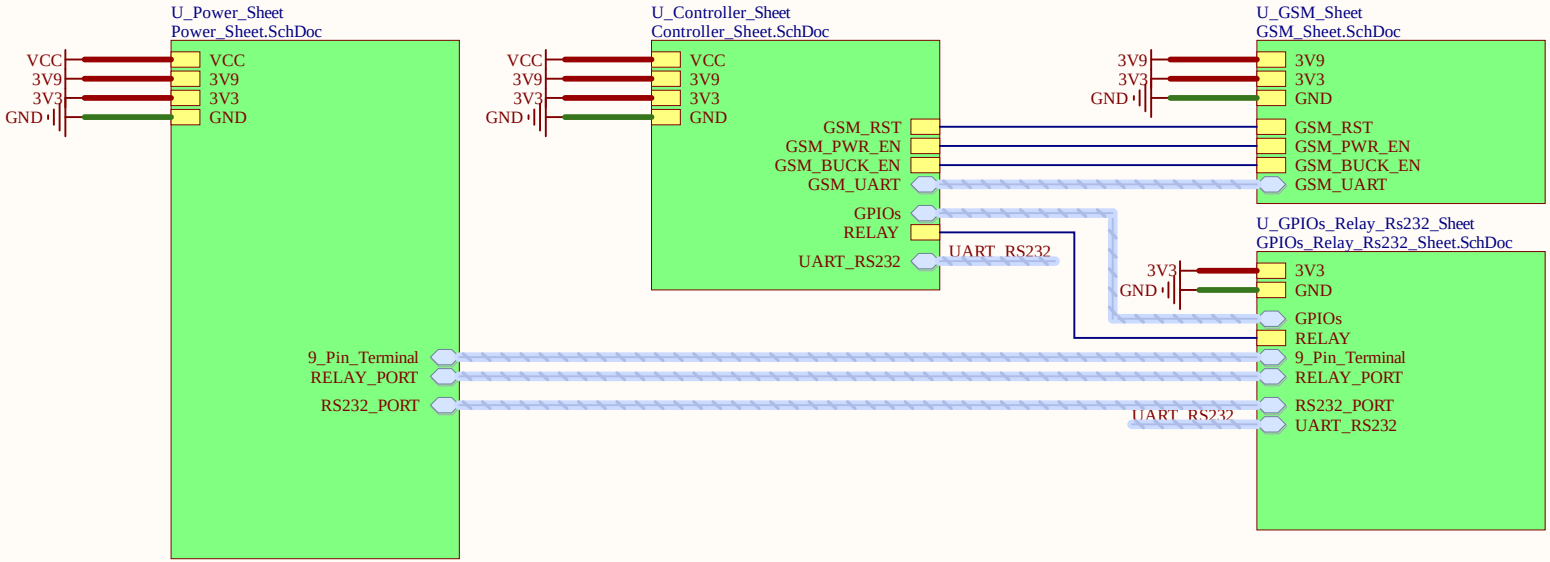
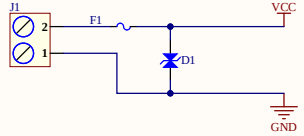


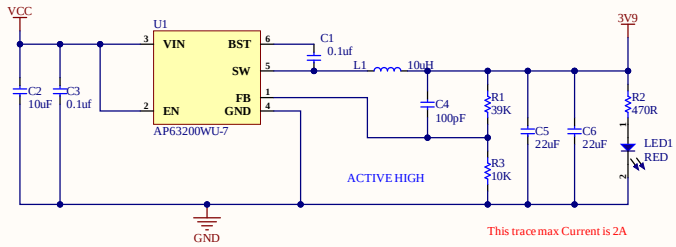
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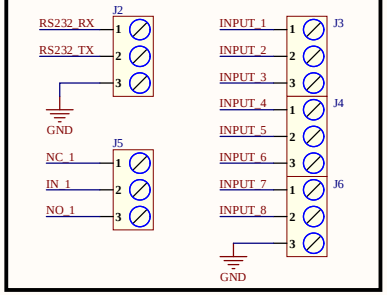
9-30V DC INPUT



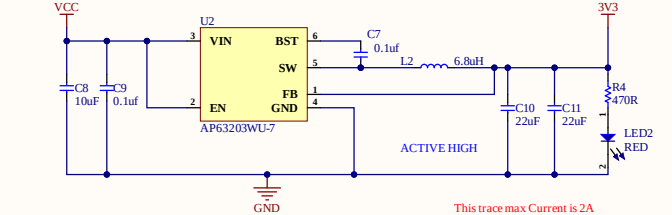
3.9V BUCK REGULATOR

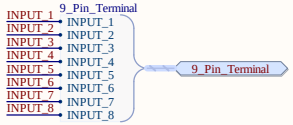


SCREW TERMINALS



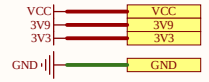
3V3 BUCK REGULATOR



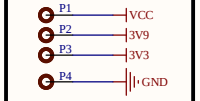








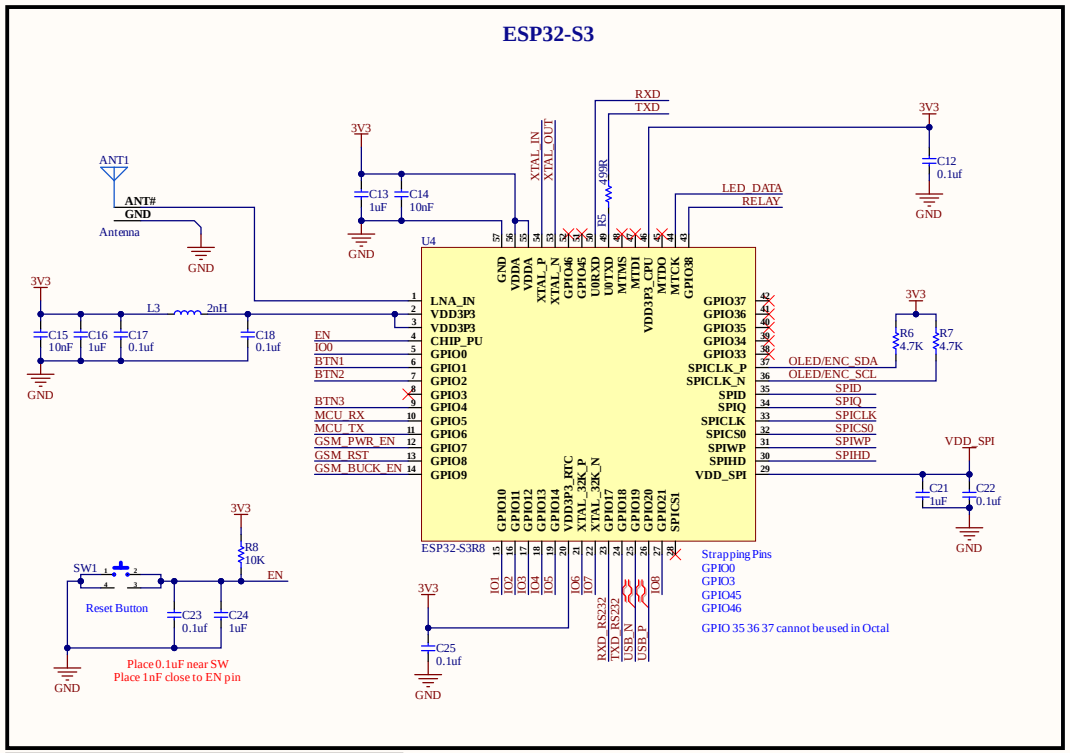
TEST POINTS



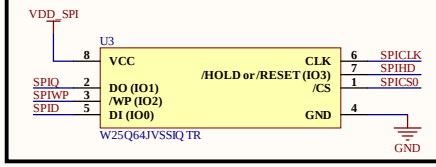
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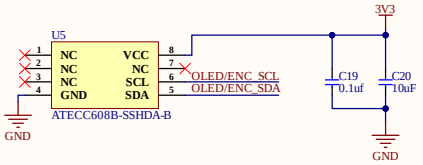
ESP32-S3



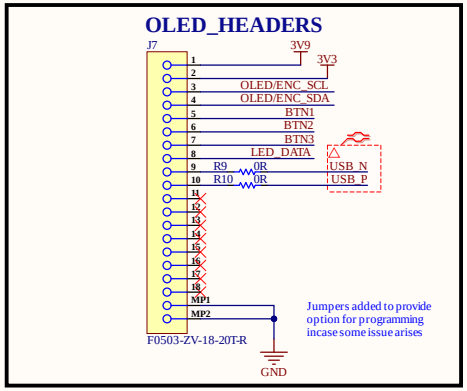
8MB FLASH



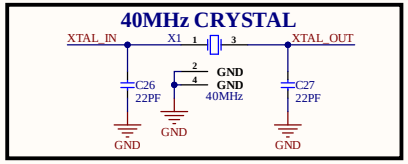
SECURE ELEMENT IC



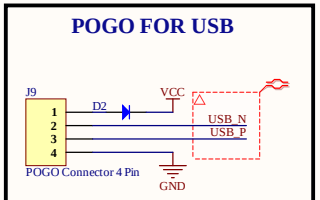
OLED_HEADERS



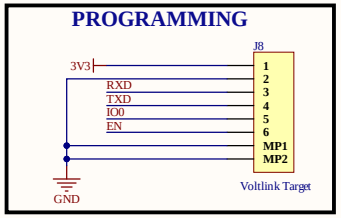
40MHz CRYSTAL



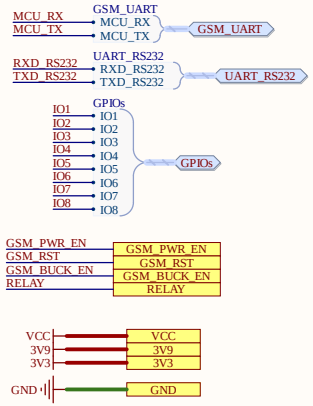
POGO FOR USB



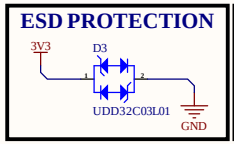
PROGRAMMING



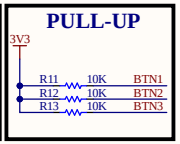
ESP32-S3 Connections



ESD PROTECTION



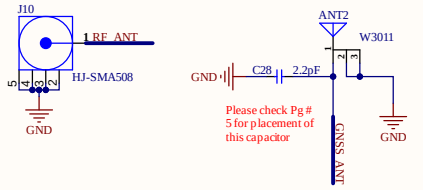
PULL-UP



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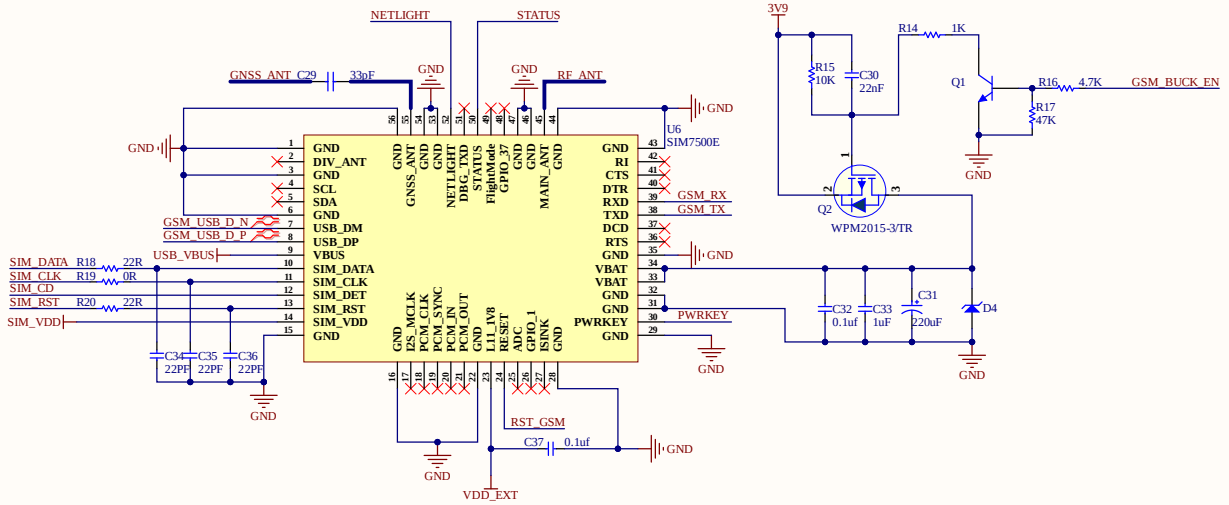
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ANTENNA



Ground cleared under antenna, clearance area 4.00 x 4.25 mm
Feed line should be designed to match 50 ohm characteristic impedance, depending on PWB material and thickness.
Recommended test board layout for electrical characteristic measurement, test board outline size 80 x 37 mm.
For more info check datasheet

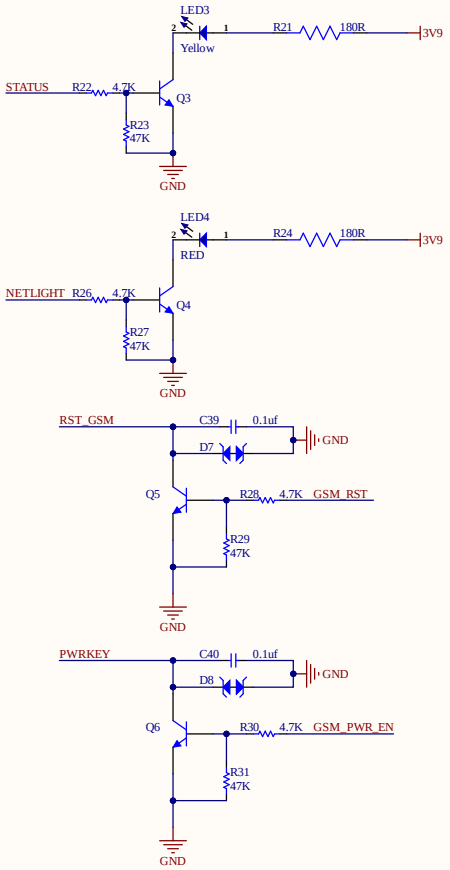
SIM7500E



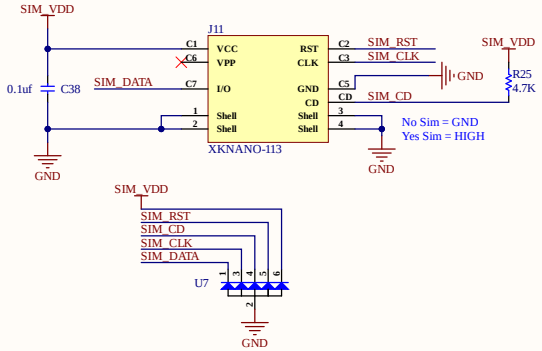
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

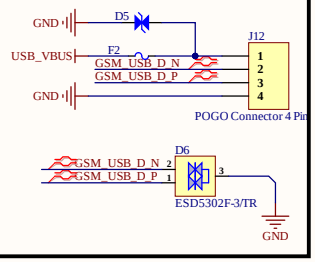
GSM_COMMANDS



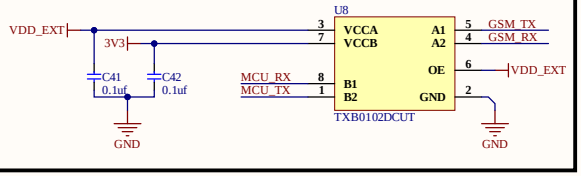
NANOSIM CONNECTOR

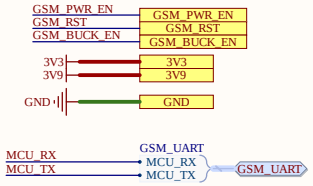


USB POGO for Updates



Level Shifter for GSM Serial

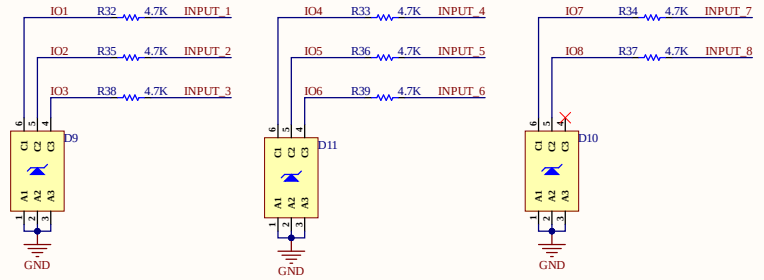




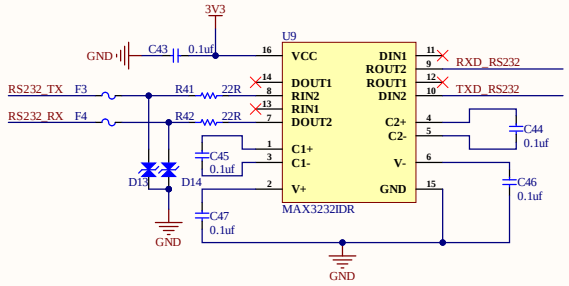
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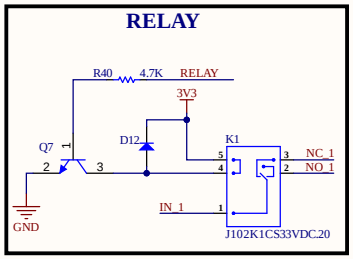
OPTO ISOLATION FOR GPIOs



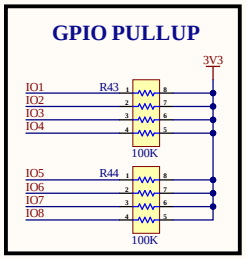
RS232 to UART

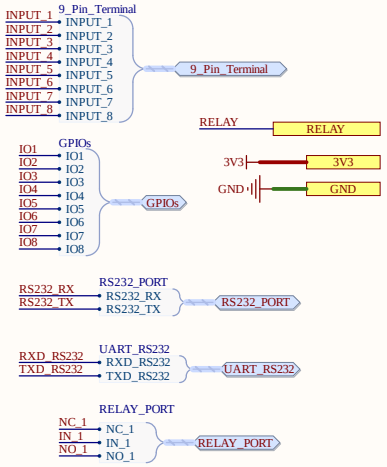


RELAY



GPIO PULLUP





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